

Visualizing the Avant-Garde: Data-Driven Insights into Artist Connections

Luisa Müller

Bachelor Thesis Colloquium

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2. Evaluated by: Jun.-Prof. Dr. Andreas Spitz

Supervisor: Raphael Buchmüller

Chair of Data Analysis & Visualization



Explosion of Diverse Artistic Movements

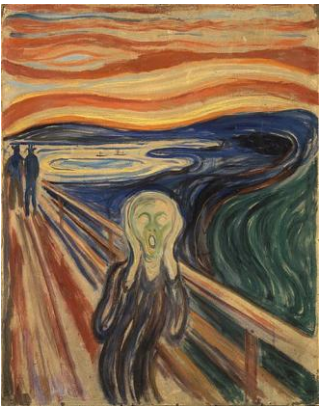
Motivation

[1] The City Rises, 1910
by Umberto Boccioni
(Futurism)

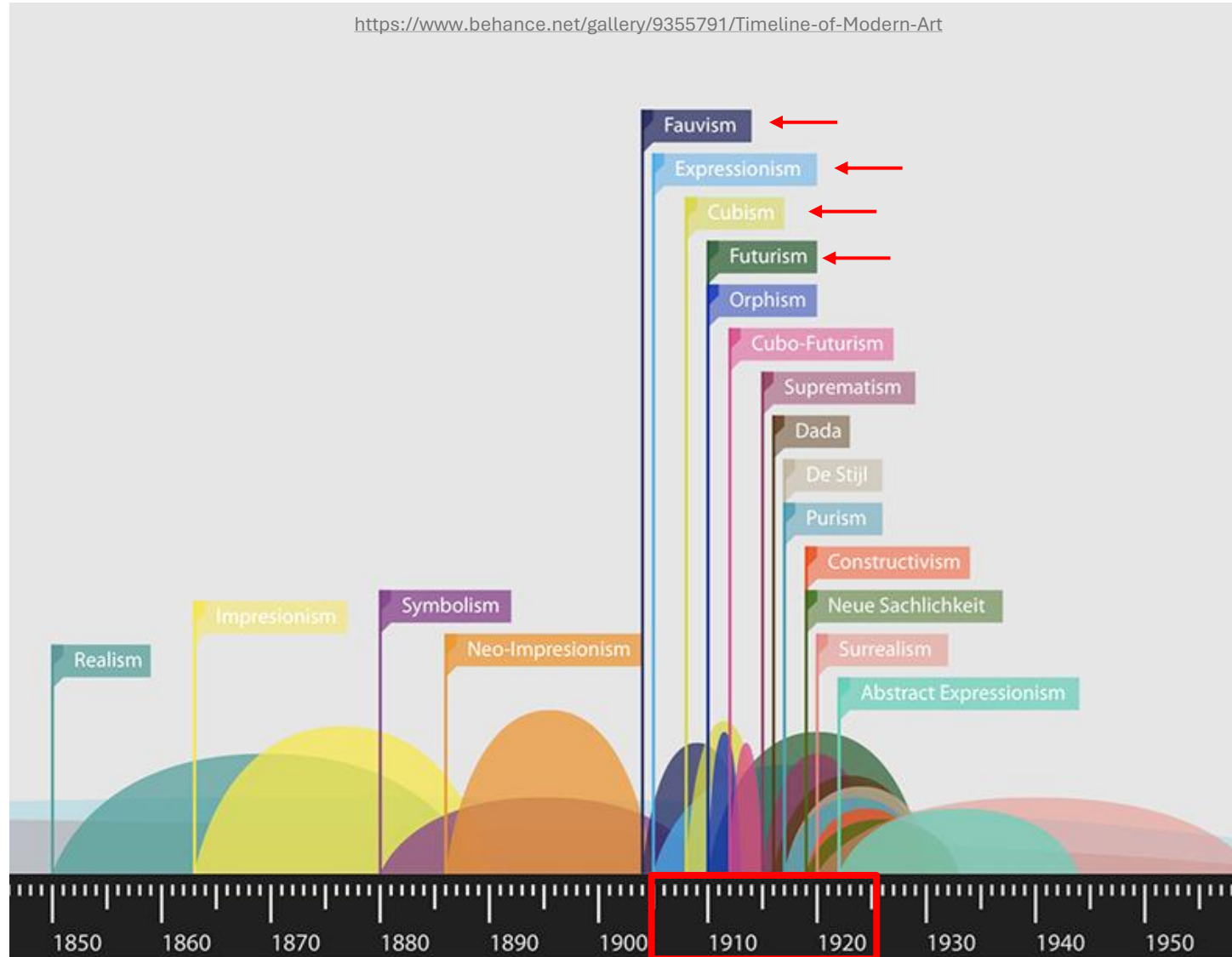


<https://www.moma.org/collection/works/79865>

[2] The Scream, 1893 by
Edvard Munch
(Expressionism)



https://www.edvardmunch.org/the-scream.jsp#google_vignette



[3] Joy of Life, 1905
by Henri Matisse
(Fauvism)



<https://www.henrimatisse.org/joy-of-life.jsp>

[4] Girl with Mandolin, 1910
by Pablo Picasso
(Cubism)



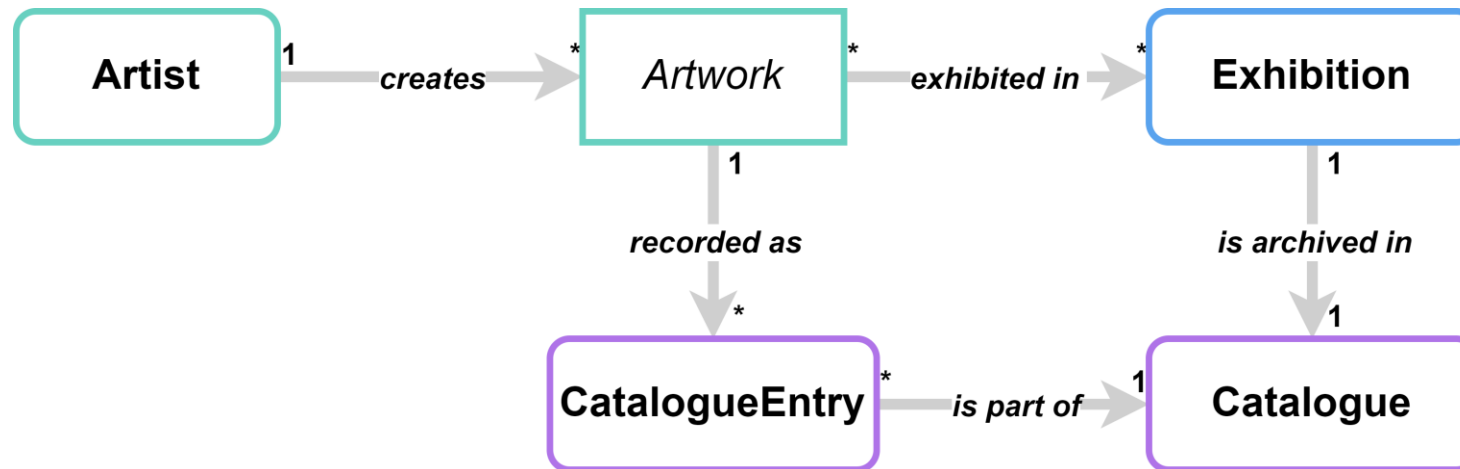
<https://www.pablocicasso.org/cubism.jsp>

Exhibitions as Hubs for Artistic Exchange

Motivation



Overview Database „DoME“



- Principal Investigator: **Raphael Rosenberg**
 - professor at the Department of **Art History** (University of Vienna)
- **European Paintings and Drawings 1905-1915**

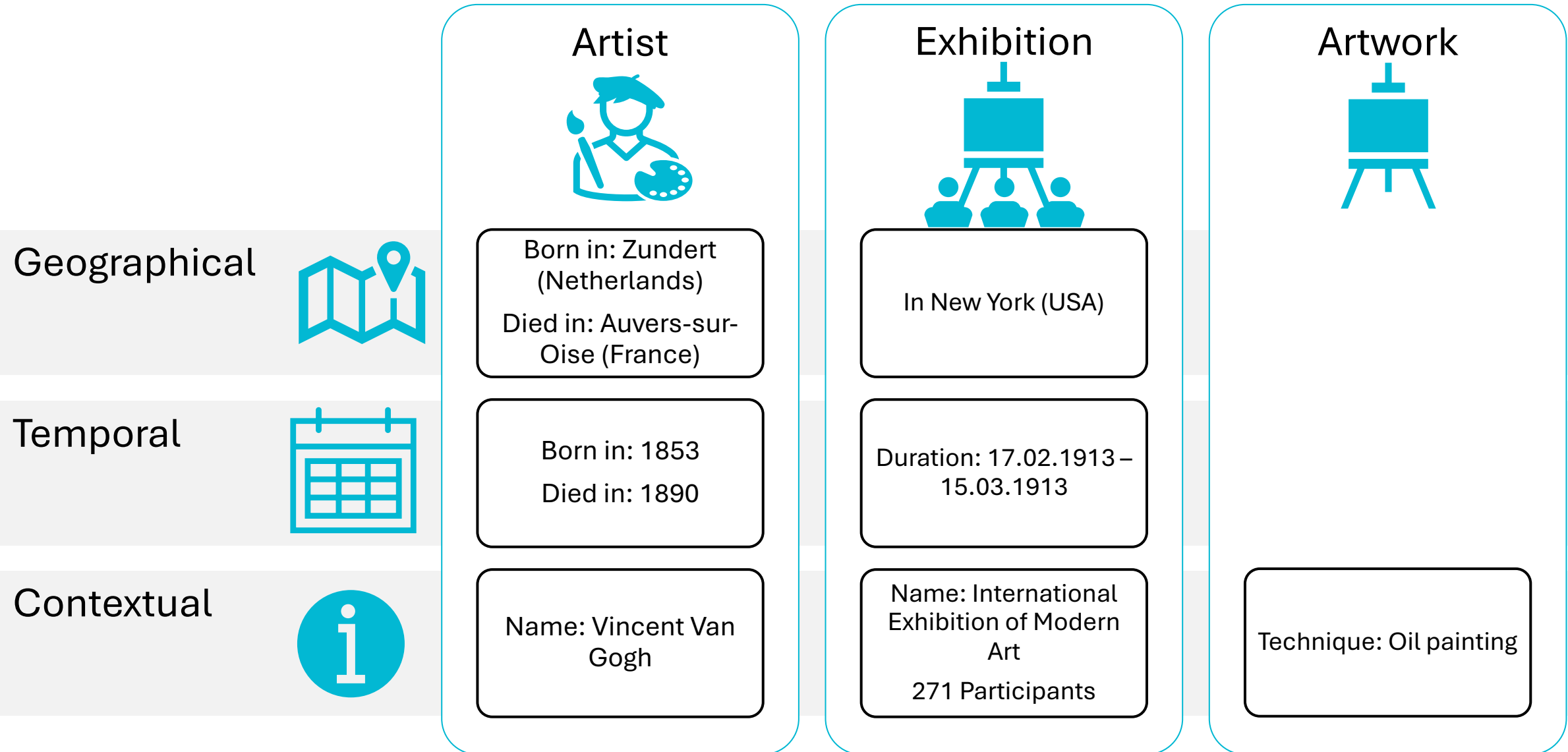
Research Question

Introduction

**How can we deepen the
Understanding of Artist Connections
through Visual Analytics?**

DoME Offers Multiple Dimensions

Introduction



Overview of Related Work

Background

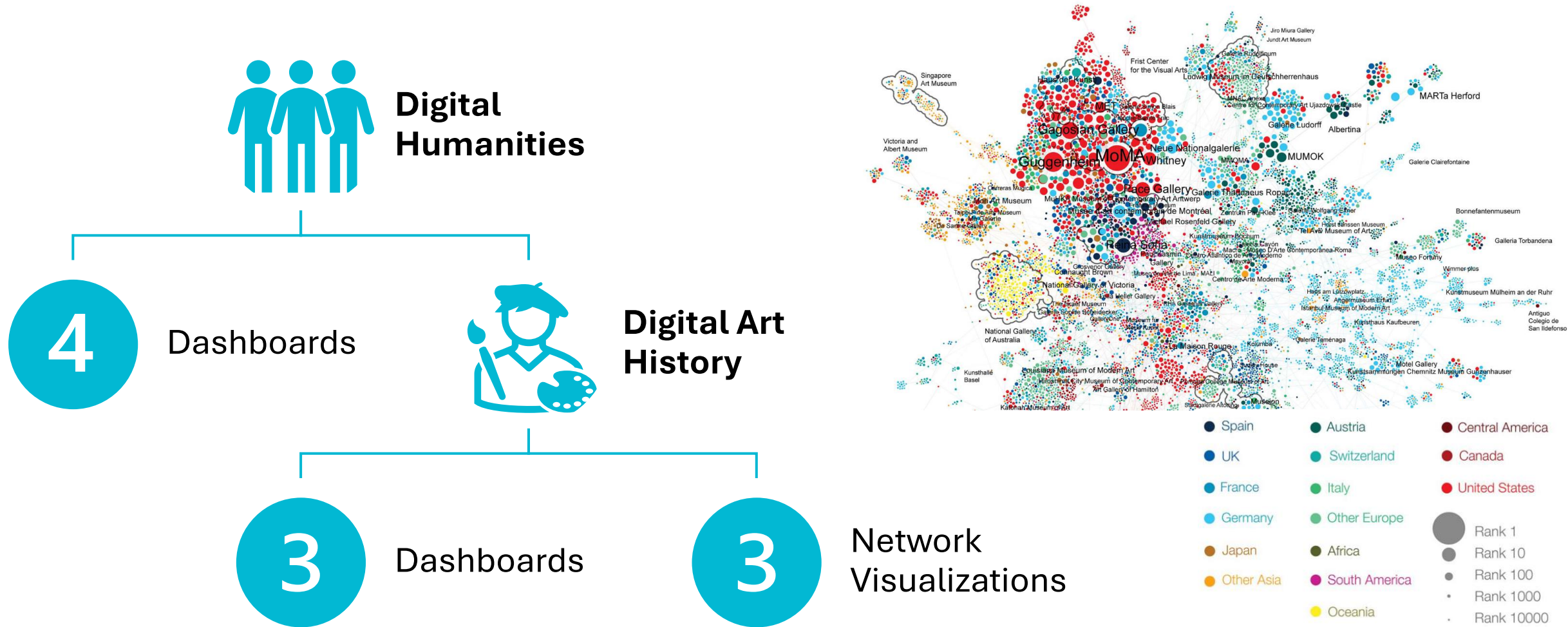


Figure 1 of Fraiberger, S. P., Sinatra, R., Resch, M., Riedl, C., & Barabási, A.-L. (2018). Quantifying reputation and success in art. *Science*, 362(6416), 825-829. <https://doi.org/10.1126/science.aau7224>

Related Work: Eurovis 2024

Background

Mapping the Avantgarde: Visualizing Modern Artists' Exhibition Activity.

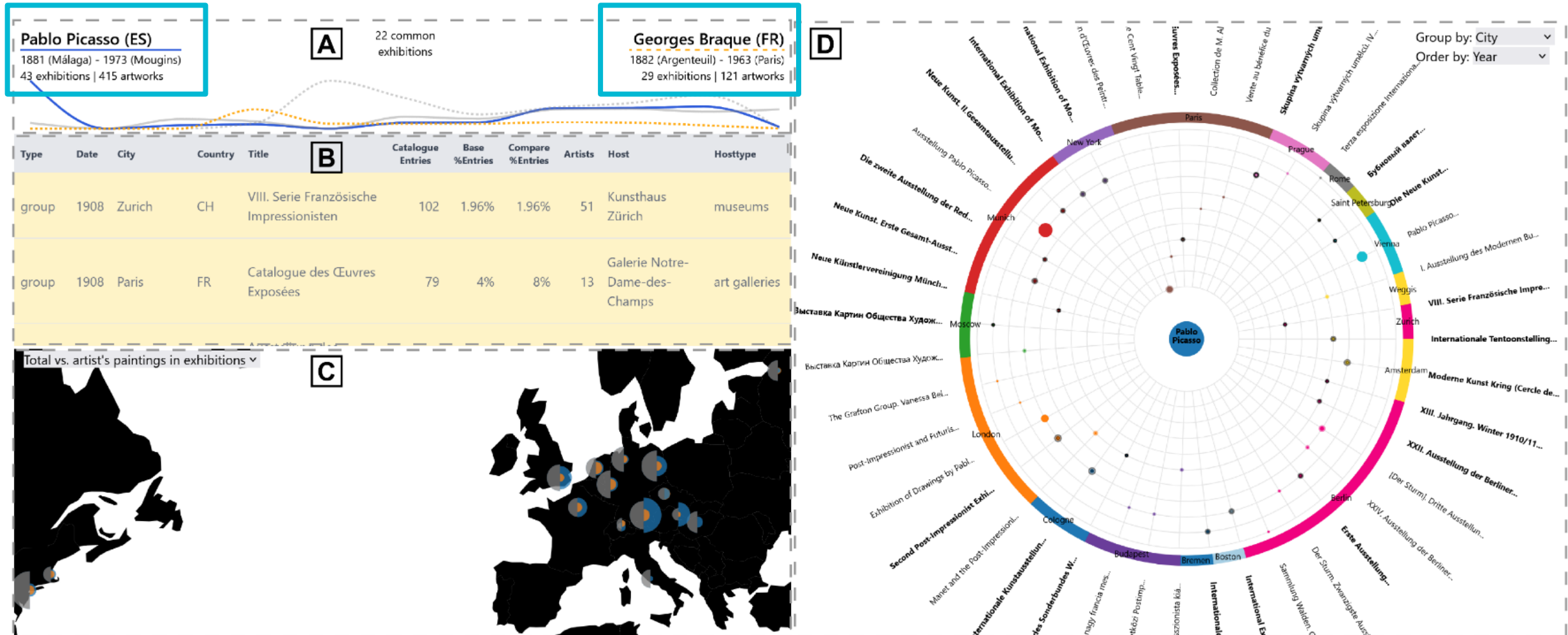


Figure 1 of Michaela Tuscher, Velitchko Andreev Filipov, Teresa Kamencek, **Raphael Rosenberg**, and **Silvia Miksch**. 2024. Mapping the Avantgarde: Visualizing Modern Artists' Exhibition Activity. *EuroVis 2024 - Short Papers* (2024). <https://doi.org/10.2312/EVS.20241063>

4 Main Analytical Tasks

Use of **Intermediate Feedback** with Research Group in Vienna
including **Raphael Rosenberg** and **Silvia Miksch**



[T1] Unveiling Artist **Networks**

- [R1.1] Show Clusters and Artists
- [R1.2] Display Intra-Cluster Connections
- [R1.3] Display Inter-Cluster Connections
- [R1.4] Highlight Influential Artists
- [R1.5] Support Interactive Exploration



[T2] Mapping **Geographic** Influences

- [R2.1] Integrate Geographic Information into Network
- [R2.2] Provide Exhibition Locations
- [R2.3] Visualize Artists' Lifespan Locations
- [R2.4] Use Maps as Visual Legends



[T3] Exploring **Temporal** Overlaps

- [R3.1] Display Artist Lifespans
- [R3.2] Provide Overview of Exhibition Periods
- [R3.3] Visualize Detailed Exhibition Timelines
- [R3.4] Facilitate Comparing Temporal Patterns

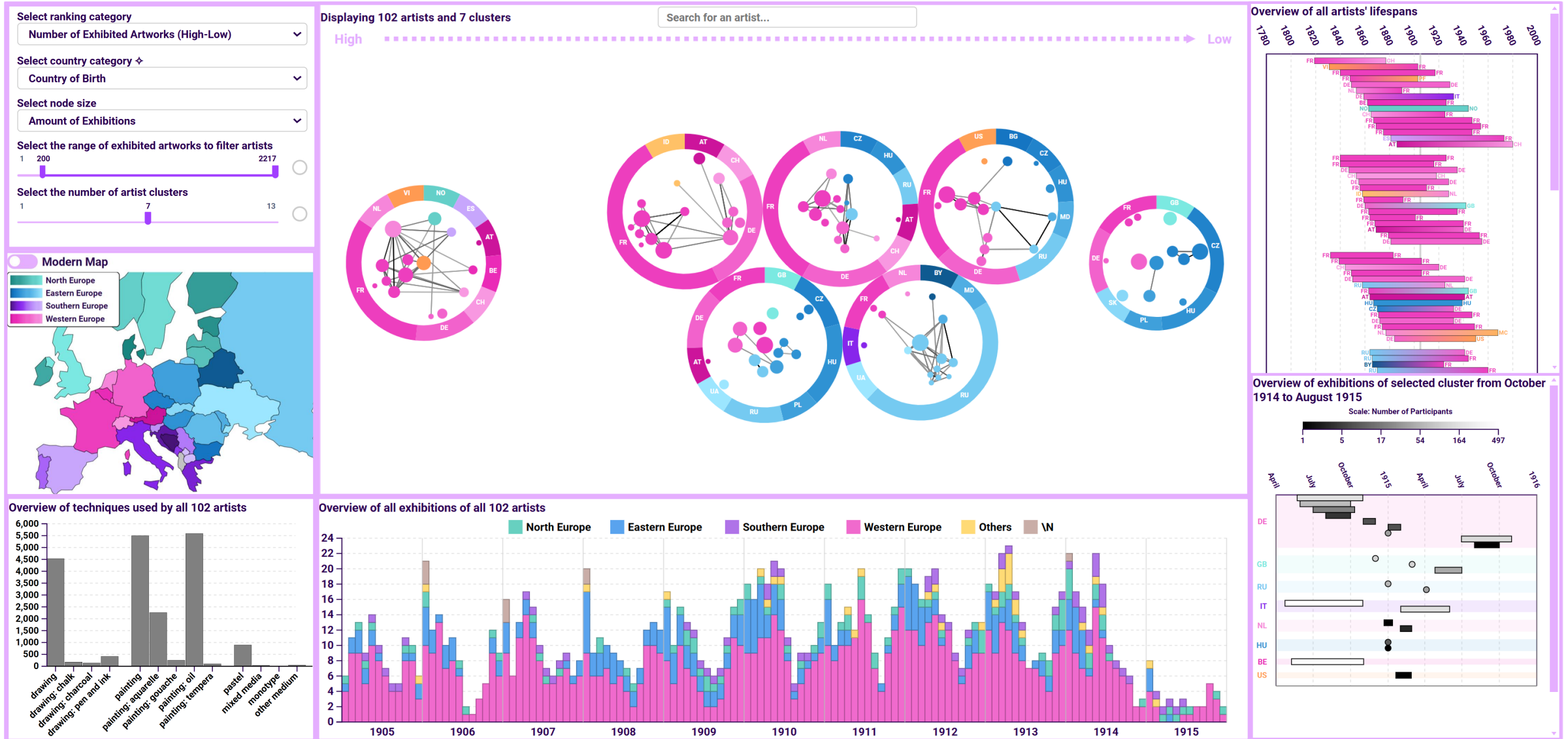


[T4] Providing Comprehensive **Context**

- [R4.1] Include Biographical Details
- [R4.2] Display Exhibition Details
- [R4.3] Provide Artist Techniques
- [R4.4] Support **AI-Generated Insights**

Overview of Dashboard

ArtistryConnect



Custom Radial Layout

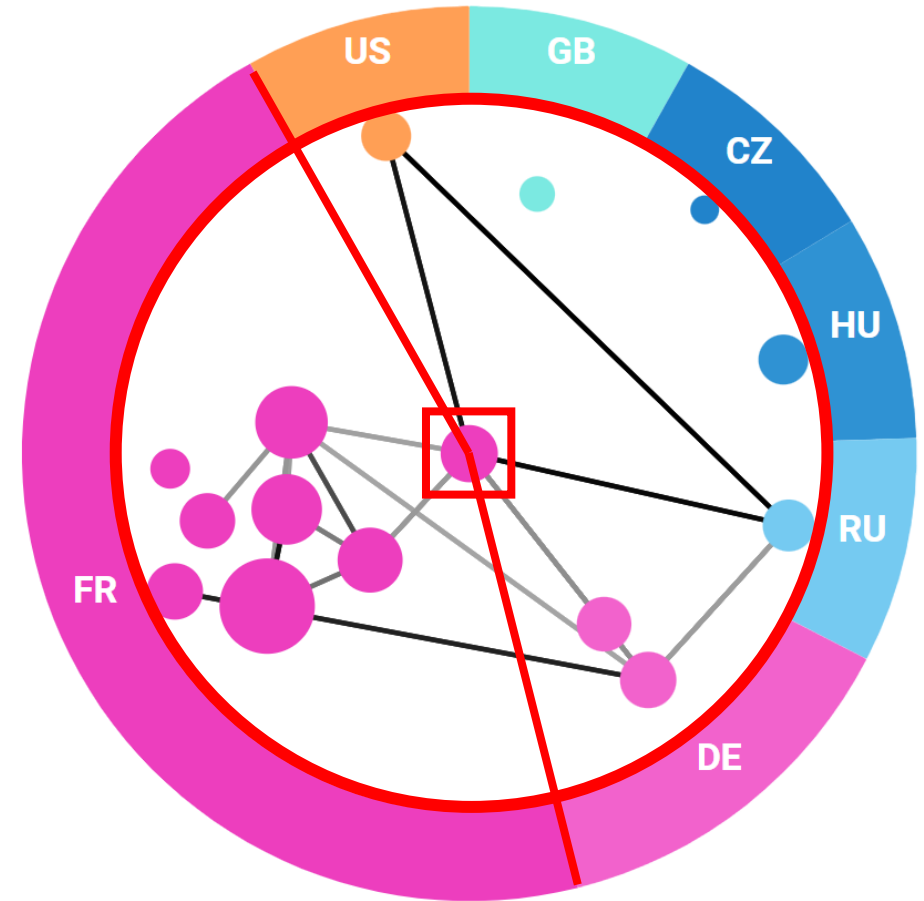
ArtistryConnect

Node - Artist

- Placed from ***inner to outer circle*** based on degree to visualize **relevance**
- Positioned within ***angle*** of (here:) their death **country**
- **Size**: Amount of Exhibitions
 - Adjustable in tool

Edges – Co-Exhibition

- Edge **weight**: based on min. number of artworks both artists exhibited
- Avoid Clutter: **threshold & color scale**



[R1.4] Highlight Influential Artists
[R2.1] Integrate Geographic Information into Network

Custom Radial Layout

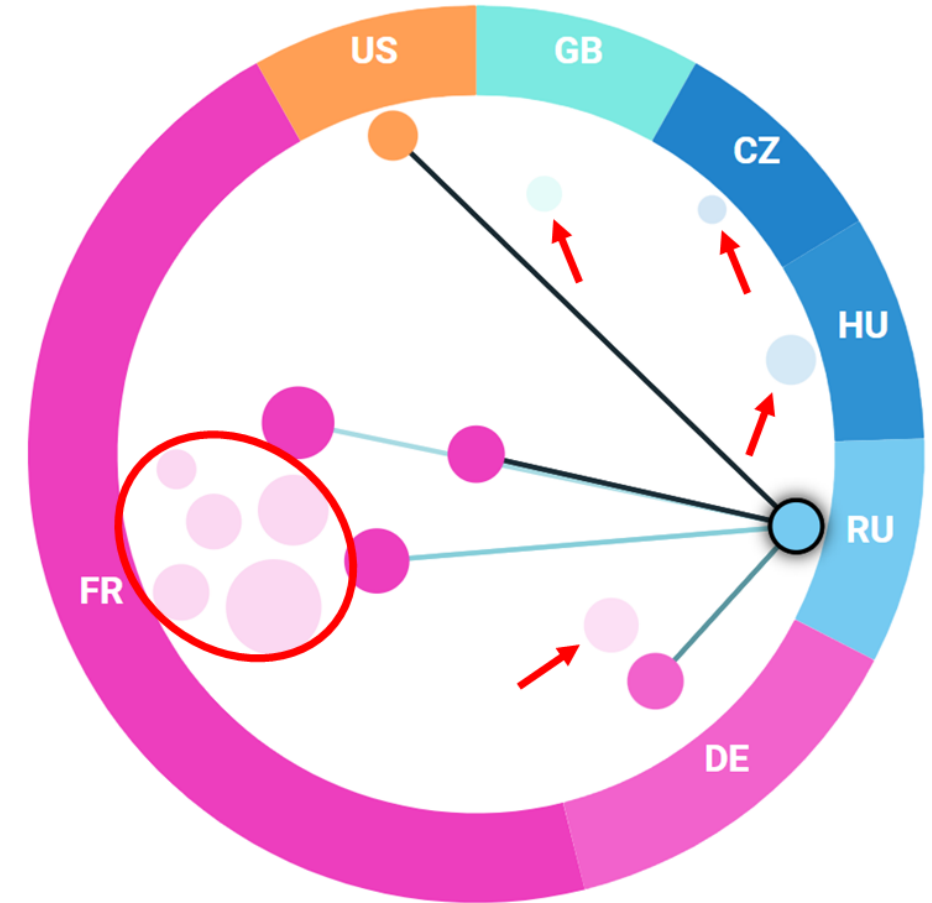
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Edges – Co-Exhibition

- Edge **weight**: based on min. number of artworks both artists exhibited
- Avoid Clutter: **threshold** & **color scale**
- **Edges visible by interaction**



- [R1.4] Highlight Influential Artists
- [R2.1] Integrate Geographic Information into Network
- [R1.2] Display Intra-Cluster Connections

Custom Clustering Approach

ArtistryConnect

- Experimented with benchmark Algorithms:
 - *Modularity Optimization, SCC, WCC, Louvain*
 - Too many clusters even for a smaller number of artists
 - **No control over the number of clusters**
- Own Implementation of **Spectral Clustering**
 - Suitable for graph structure
 - Utilizes edge weights

Main Steps:

1. Adjacency & Degree Matrix Construction
2. Laplacian Matrix Calculation & Normalization
3. Eigenvalue Decomposition
4. **K-means Clustering**

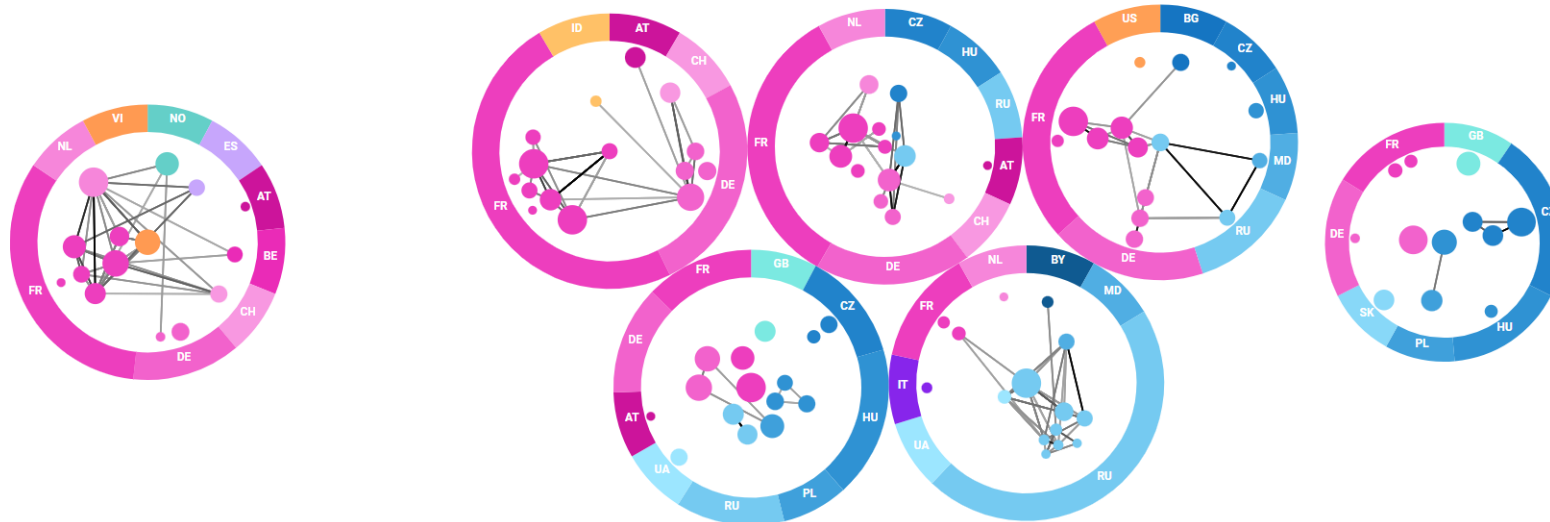
ArtistryConnect



Search for an artist...

High

Low

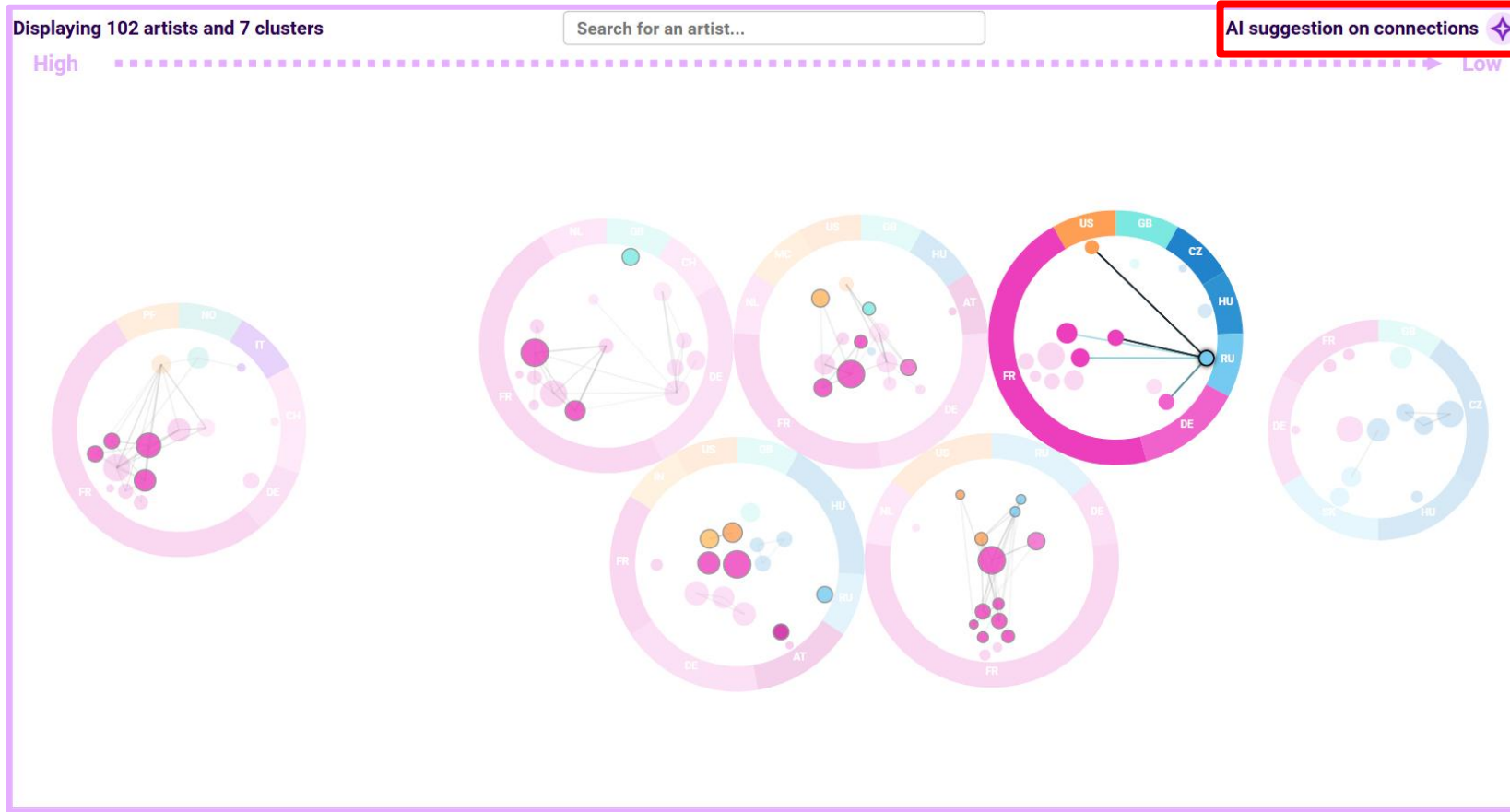


[R1.1] Show Clusters and Artists



Inter-community Connections

ArtistryConnect



[R4.4] Support AI-Generated Insights

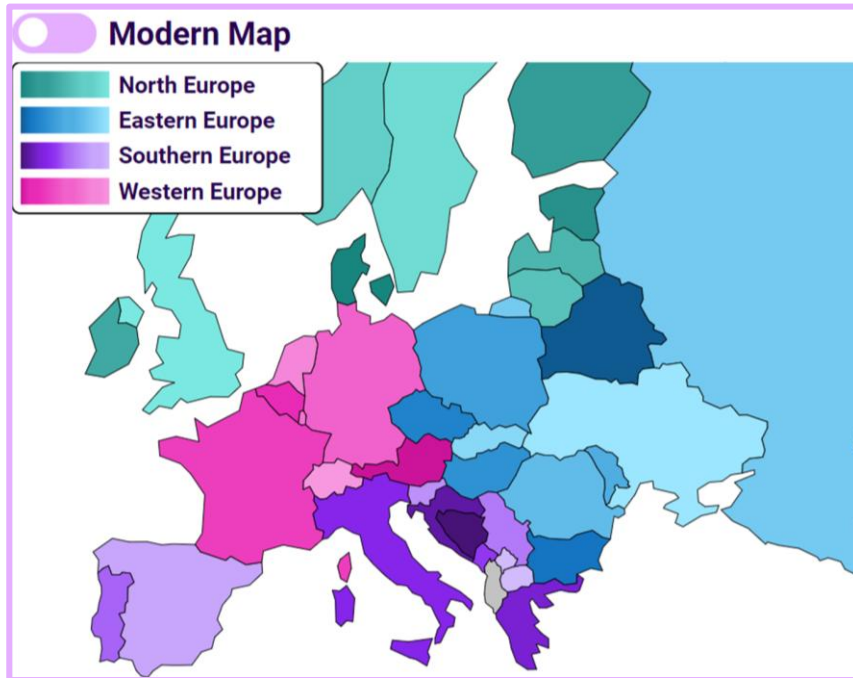


[R1.3] Display Inter-Cluster Connections



Regional Color Schemes

ArtistryConnect

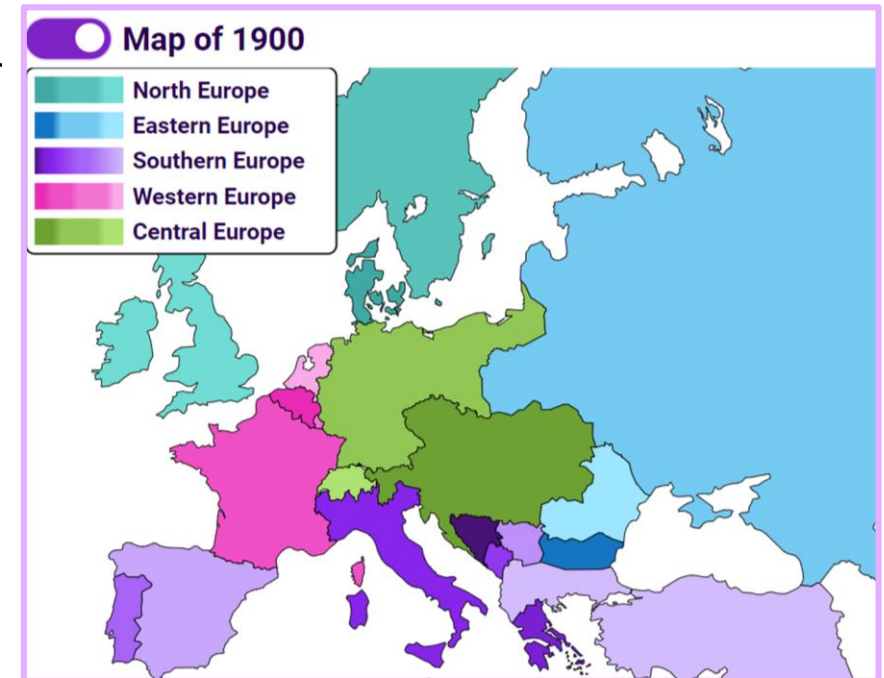


Goal: Support Geographical tasks in whole dashboard

For non-european countries:



Split of Regions: Geoscheme of United Nations Statistics Division (**UNSD**)



Central Europe added based on **Raphael Rosenberg's** feedback

(intermediate feedback)

[R2.4] Use Maps as Visual Legends

UNSD: <https://unstats.un.org/unsd/methodology/m49/>

Navigation Component

ArtistryConnect



[R1.5] Support Interactive
Exploration

Select ranking category

Number of Exhibited Artworks (High-Low) ▼

Select country category ✦

Country of Birth ▼

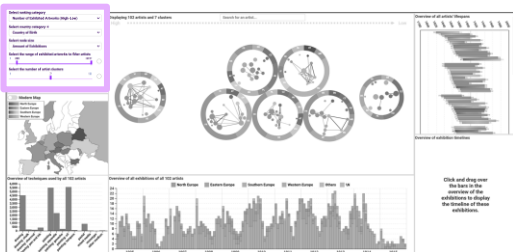
Select node size

Amount of Exhibitions ▼

Select the range of exhibited artworks to filter artists



Select the number of artist clusters



Navigation Component

ArtistryConnect



[R1.5] Support Interactive
Exploration

Select ranking category

Number of Exhibited Artworks (High-Low) ▼

Select country category ✧

Country of Birth ▼

Select node size

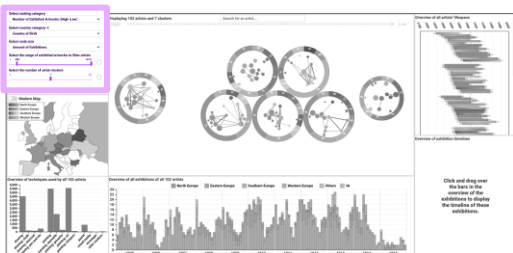
Variety of Used Techniques ▼

Select the range of exhibited artworks to filter artists

1 200 2217

Select the number of artist clusters

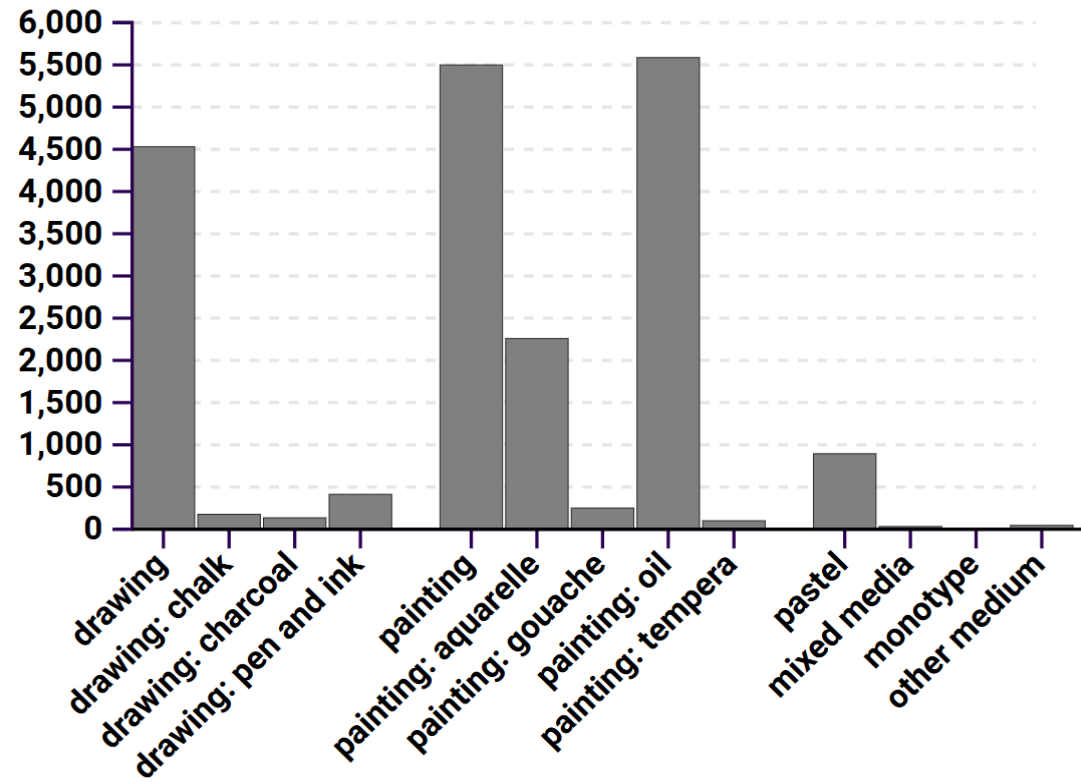
1 7 13



Techniques

ArtistryConnect

Overview of techniques used by all 102 artists



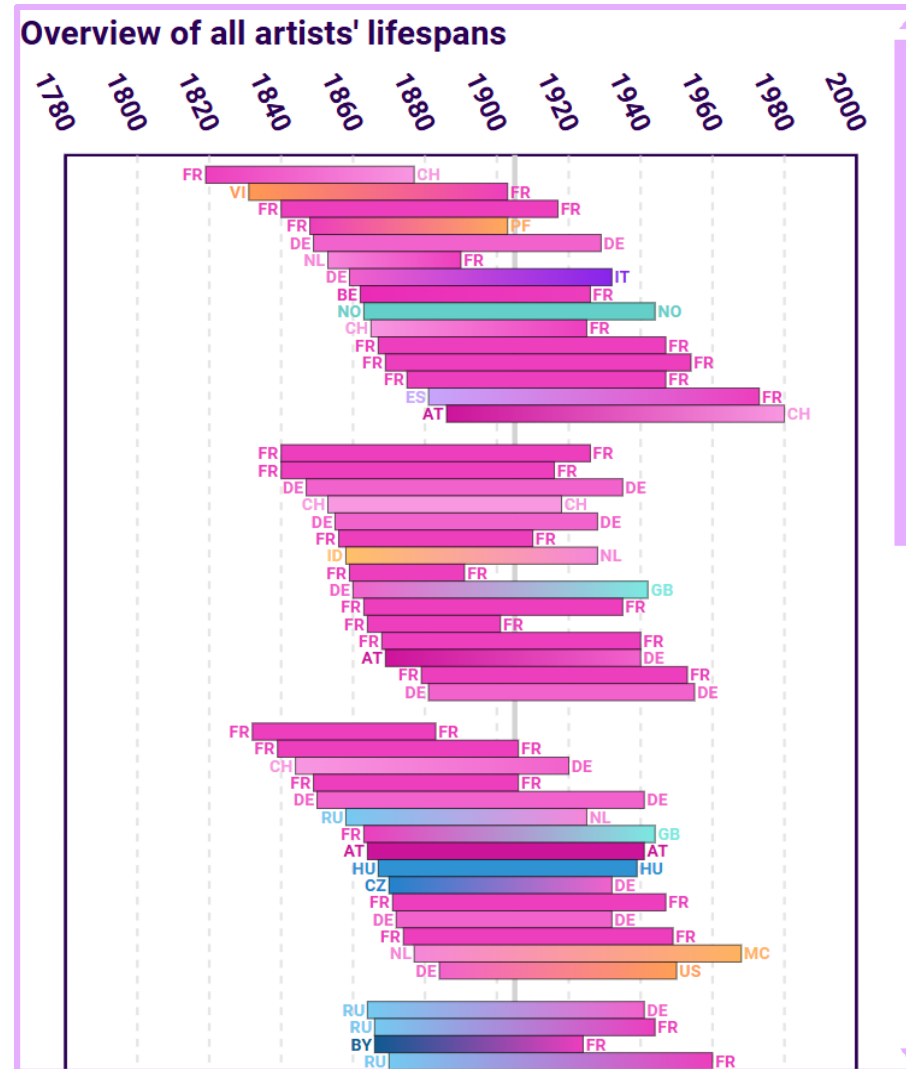
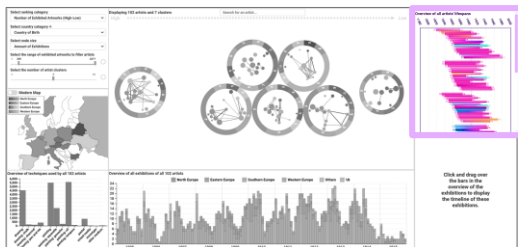
[R4.3] Provide Artist Techniques

Artist Lifespans

ArtistryConnect

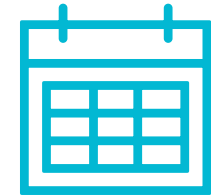


[R4.1] Include Biographical Details



[R2.3] Visualize Artists' Lifespan

Locations

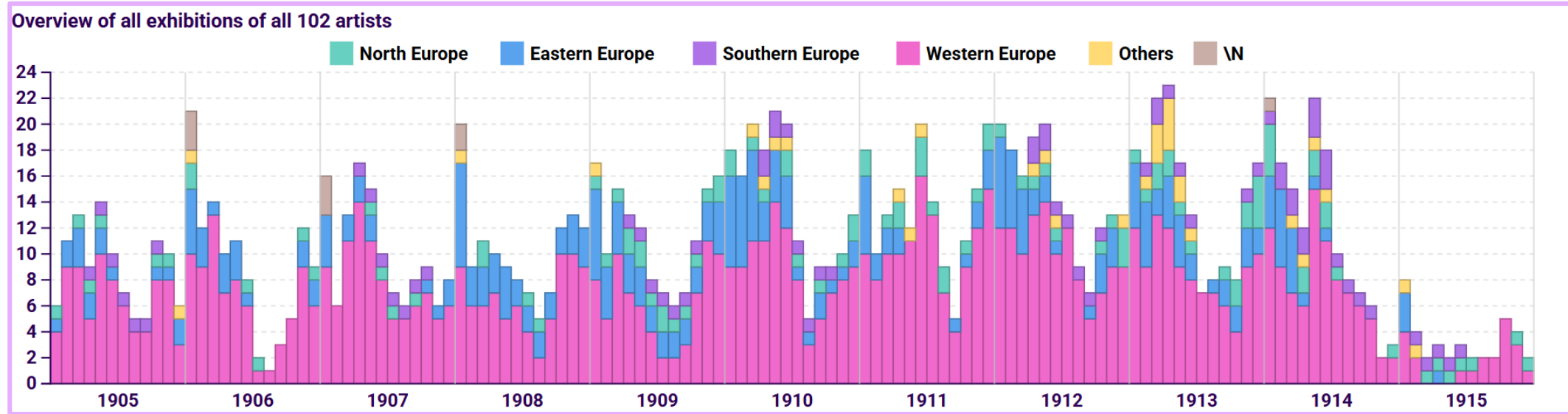


[R3.1] Display Artist Lifespans

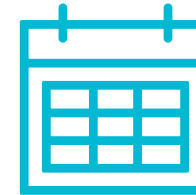
[R3.4] Facilitate Comparing Temporal
Patterns

Exhibitions

ArtistryConnect



[R2.2] Provide Exhibition Locations



[R3.2] Provide Overview of Exhibition Periods

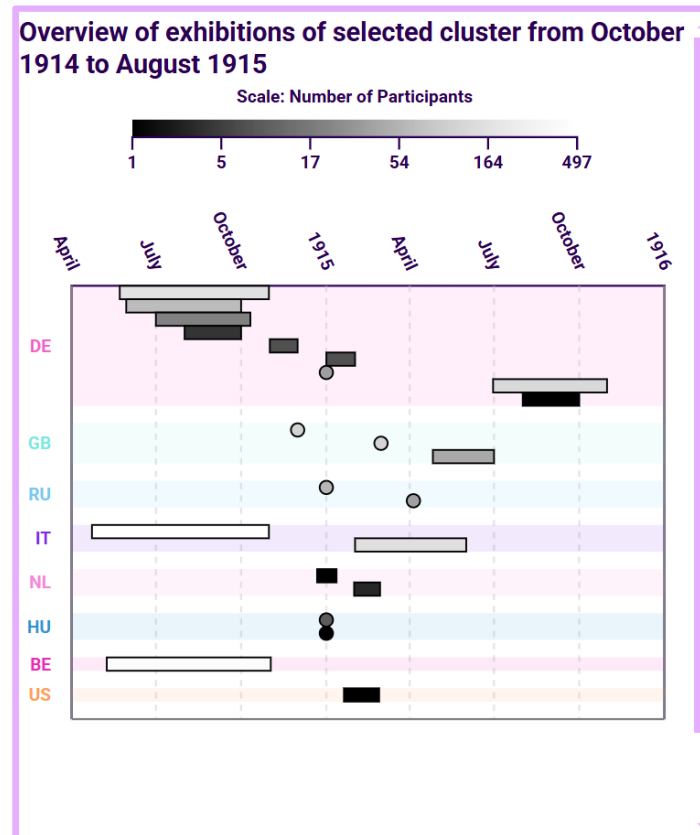
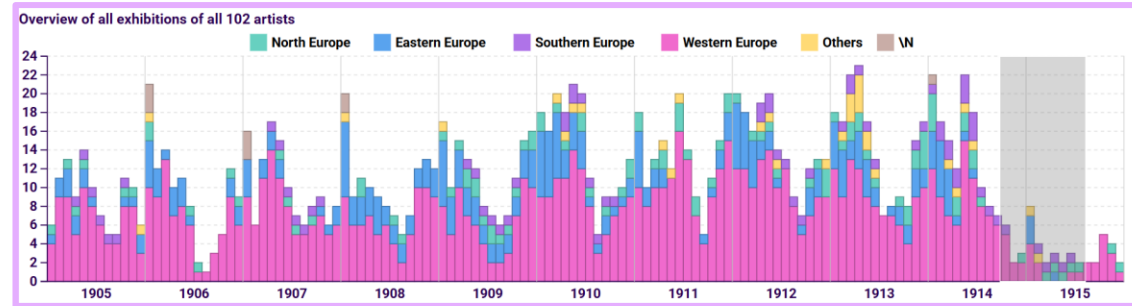
[R3.4] Facilitate Comparing Temporal Patterns

Details of Exhibitions

ArtistryConnect



[R4.2] Display Exhibition Details



[R2.2] Provide Exhibition Locations



[R3.3] Visualize Detailed Exhibition

Timelines

[R3.4] Facilitate Comparing Temporal

Patterns

Live-Demo

ArtistryConnect



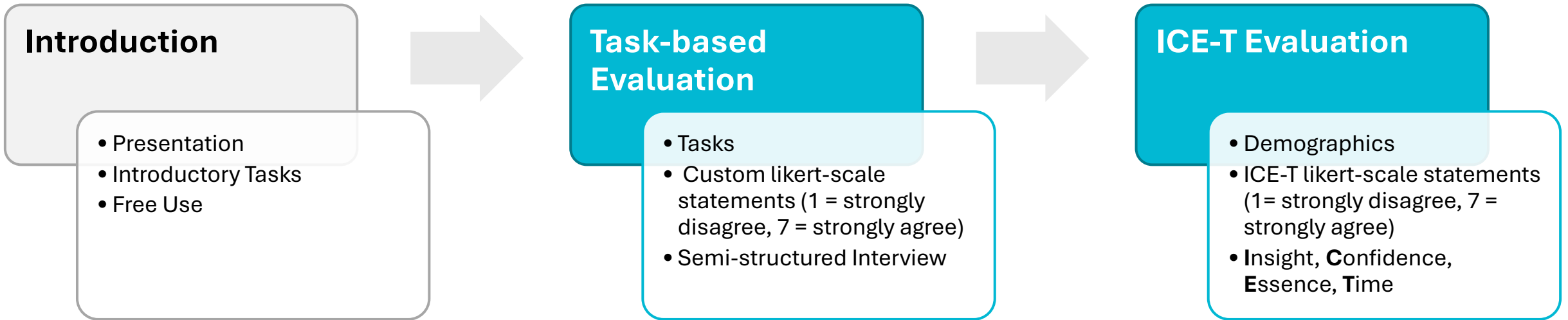
Sources validating Findings on Connections:

- “Through [Pissarro’s] dedication to the Impressionist movement, he earned the nickname “The father of Impressionism”. ” <https://artincontext.org/camille-pissarro>
- John Rewald. 1938. Camille Pissarro: His Work and Influence. The Burlington Magazine for Connoisseurs 72, 423 (1938), 280–291.
- <https://www.vangoghmuseum.nl/en/prints/person/39732/pissarro-camille>
- <https://www.riseart.com/article/2607/gauguin-and-van-gogh-an-explosive-friendship>
- <https://www.tricera.net/artclip/blog912>
- <https://www.kunstimunterricht.de/kunstgeschichte/stilrichtungen/72-impressionismus-allgemeine-grundlagen.html>

Study Design

Study

3 Phases

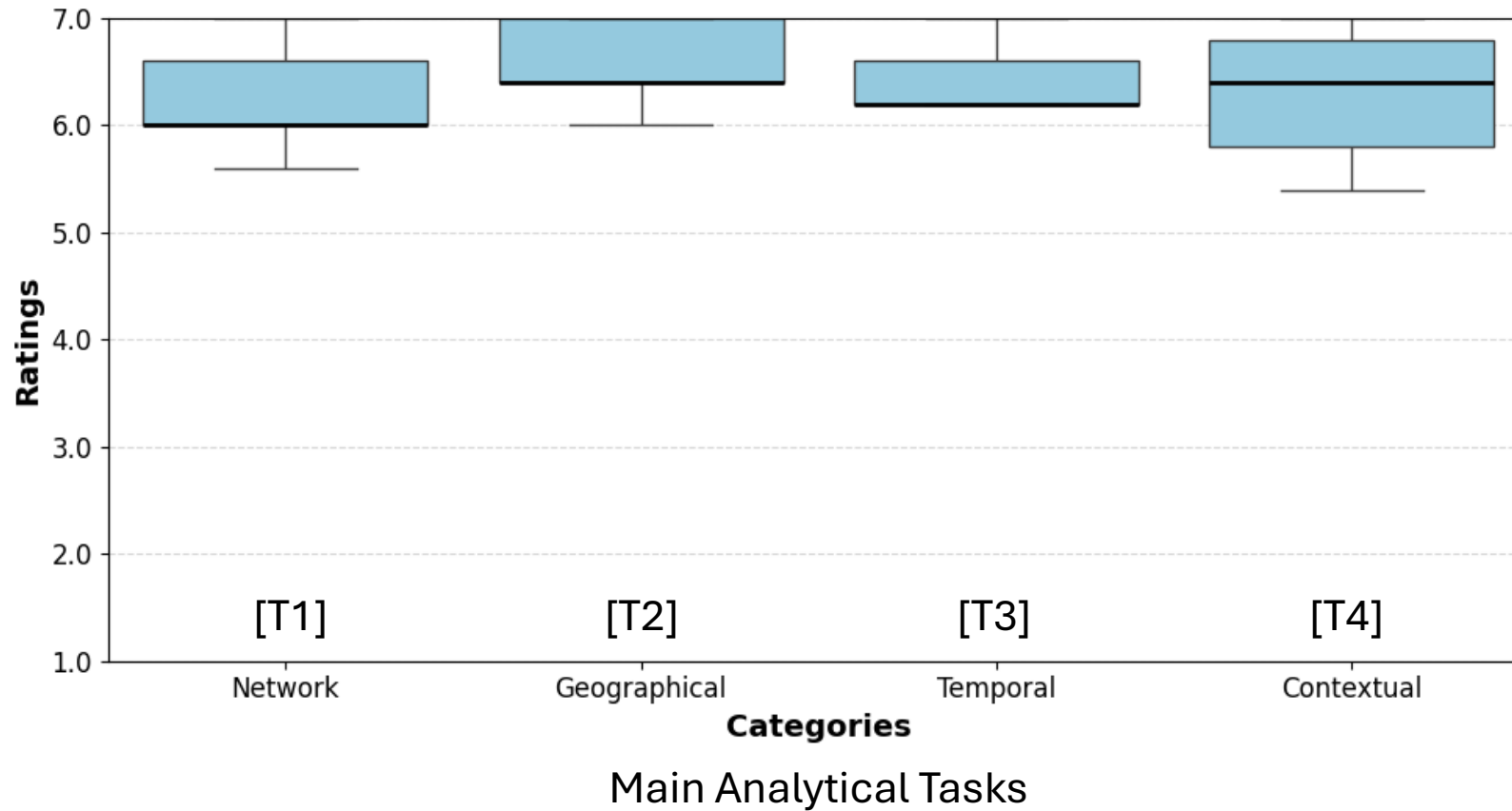


Demographics of Participants

ID	Education Level	Area of Expertise	Gender	Age
DB	Student	Familiar with Database	F	23
VA	Master's Degree	Visual Analytics Professional	F	26
AH	Master's Degree	Art Historian	F	28
AS1	Student	Art History Student	F	39
AS2	Bachelor's Degree	Art History Student	F	33

Task-based Results

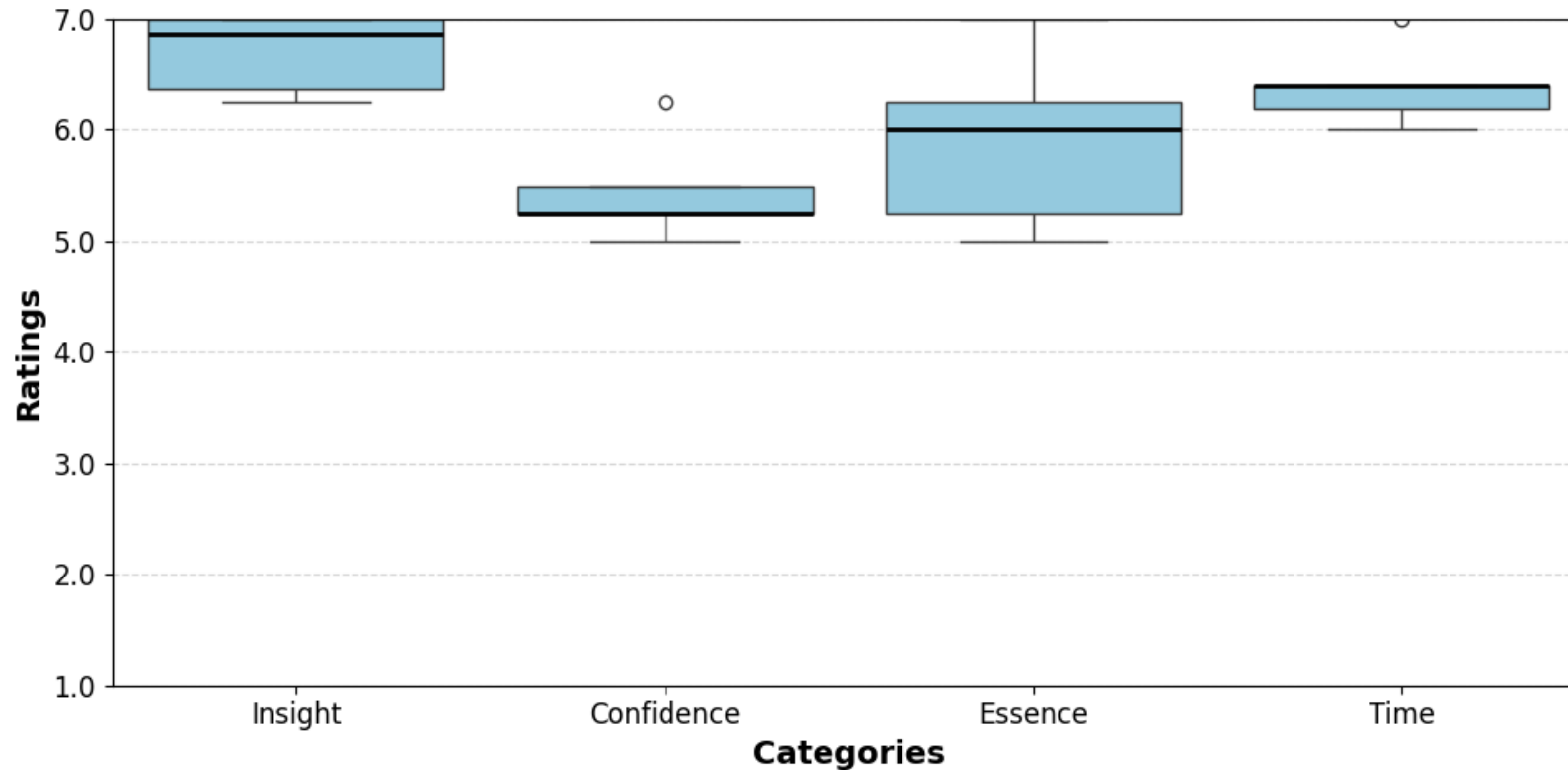
Study



- Variety in Contextual:
 - Critique on techniques
 - Doubts on AI accuracy

ICE-T Results

Study



- No indicator for **data quality** in tool
- Different Opinions on how well the tool communicates the **data essence**
- Generate **Insights** in Short **Time**

Tool Most Effective in Generating Data-Driven Insights

Discussion

Category	Guidelines	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Insight	The visualization facilitates answering questions about the data.	The visualization exposes individual data cases and their attributes.	6,6	7	7	7	5	7
		The visualization facilitates perceiving relationships in the data like patterns & distributions of the variables.	6,6	7	6	7	6	7
		The visualization promotes exploring relationships between individual data cases as well as different groupings of data cases.	6,8	7	6	7	7	7
	The visualization provides a new or better understanding of the data.	The visualization helps generate data-driven questions.	7	7	7	7	7	7
		The visualization helps identify unusual or unexpected, yet valid, data characteristics or values.	6,6	6	6	7	7	7
	The visualization provides opportunities for serendipitous discoveries	The visualization provides useful interactive capabilities to help investigate the data in multiple ways.	6,8	7	7	7	6	7
		The visualization shows multiple perspectives about the data.	6,6	7	6	7	6	7
		The visualization uses an effective representation of the data that shows related and partially related data cases.	6,6	7	6	7	6	7

„This is **interesting**“ – AH, AS1

„There are **more connections** here than i see in the database“ – DB

„I am just realizing that your tool really shows me **things that I didn't know**. And I really **want to look into that.**“ – AH

1=Strongly Disagree, 7=Strongly Agree



[T1] Interactive Exploration of Networks

Discussion

[R1.5] Support Interactive Exploration
[R1.1] Show Clusters and Artists

Positioning of clusters facilitated cross-cluster **comparisons**

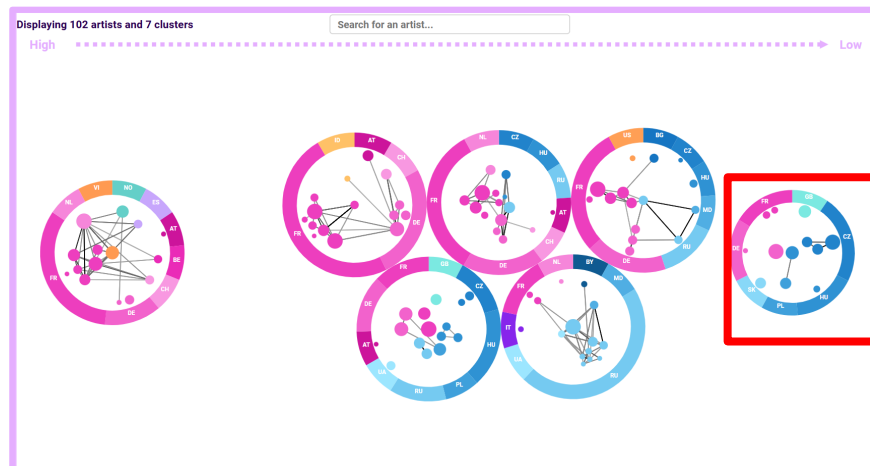
Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Network	Switching the ranking category allowed me to compare clusters and through that gain insights into the characteristics of the artist groups.	6,6	6	6	7	7	7
	The structure of a cluster made it easy to see which artists were connected.	6,6	6	7	7	7	6
	The colored lines (edges) between artists within clusters helped me easily spot strong connections.	6	5	7	7	5	6
	Switching the country category and node sizes gave me better insights into artist connections and the connected artists.	6,4	7	5	7	6	7
	It was easy to track and compare changes when I adjusted rankings, country categories, or node sizes.	5,6	6	3	7	5	7

„[Helps to understand] why these clusters even exist and how they're related to each other“ – VA

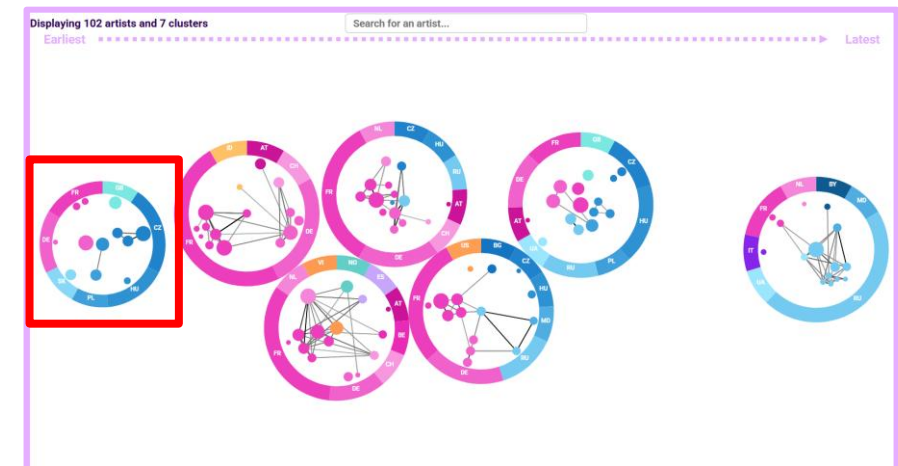
„[It serves as a] very good **comparison** and **overview**“ – AS2

„It was really **interesting** to see how [the visualization] **changes**“ – AH

Number of Exhibited Artworks



Average Death year

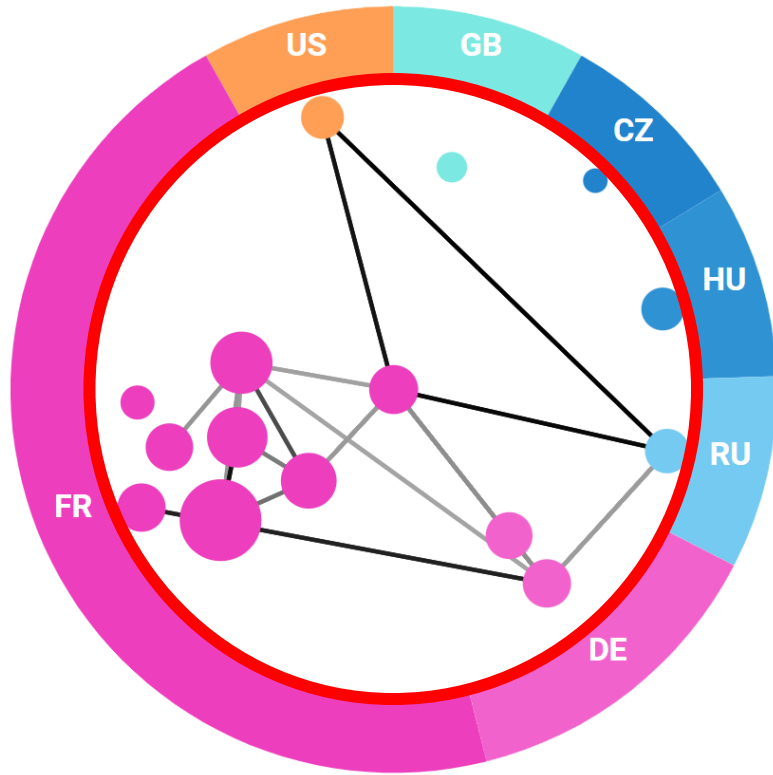




[T1] Positioning as an important visual variable

Discussion

[R1.4] Highlight Influential Artists



„a **great idea**“ for understanding each artist’s relative importance – AS2

„How they are placed allowed me to **instantly** see [**central figures**]“ - AH

„[Being able to see influential artists is] **very important**“ – AS2



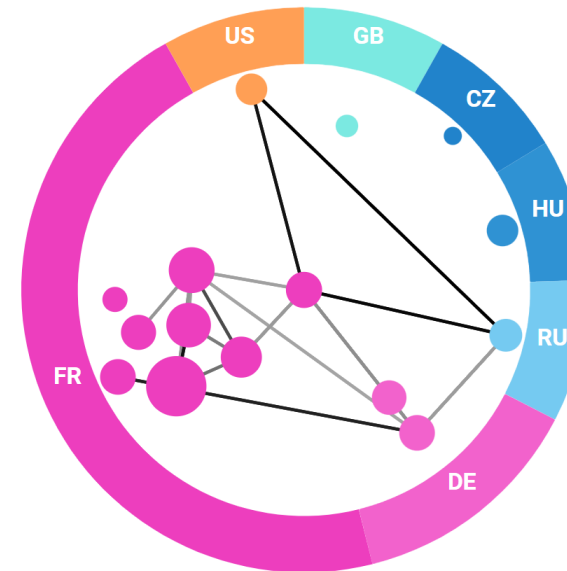
[T1] Cluster Structure Considered Useful

Discussion

[R1.2] Display Intra-Cluster Connections

Color Value for edges facilitated recognition of **strong connections**

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„You **wouldn't** know which connection is really strong **[without this visual cue]**“ – VA

Frequently used by participants to find **important and interesting connections**

⚡ Clutter & Edges cutting through nodes=> Edge Bundling?

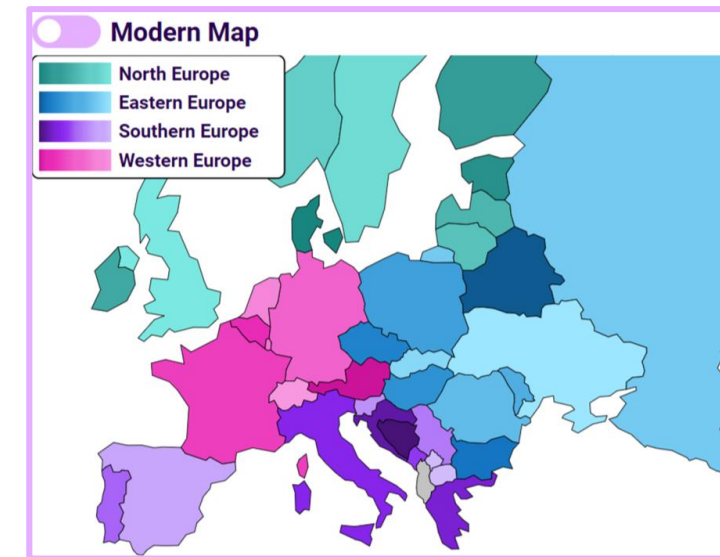


[T2] Regional Color Scale Facilitates Tasks

Discussion

[T2] Mapping **Geographic Influences**

Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Geographical	The tool made it easy to extract key geographic details about artists, like their birth, death, and exhibition locations.	7	7	7	7	7	7
	The gradient coloring of the rectangles (artist) in the artist lifespans let me quickly identify geographic patterns among artists.	6,6	7	7	7	5	7
	The cluster structure helped me identify geographical patterns in artist connections, such as artists from the same country/european region having strong connections.	6,2	5	6	7	6	7
	The overview of the exhibitions let me easily spot the most prominent european regions of exhibitions.	7	7	7	7	7	7
	The timeline of the exhibitions was useful in quickly identifying the most prominent countries of exhibitions.	6	6	5	7	5	7



„[The regional color scheme] helped because it makes it **easier to distinguish different regions**“ – DB

AH appreciated the regional color scheme for highlighting „**social and socioeconomic contexts**“

Color scale gives a „**good overview**“ and is „**aesthetic**“ – AS2

⚡ **Similarity of some colors:** „You really **have to look closely**. You can't just scan it with [the] colors“ – AS1



[T2] & [T3] Geographical and Temporal Information

Discussion

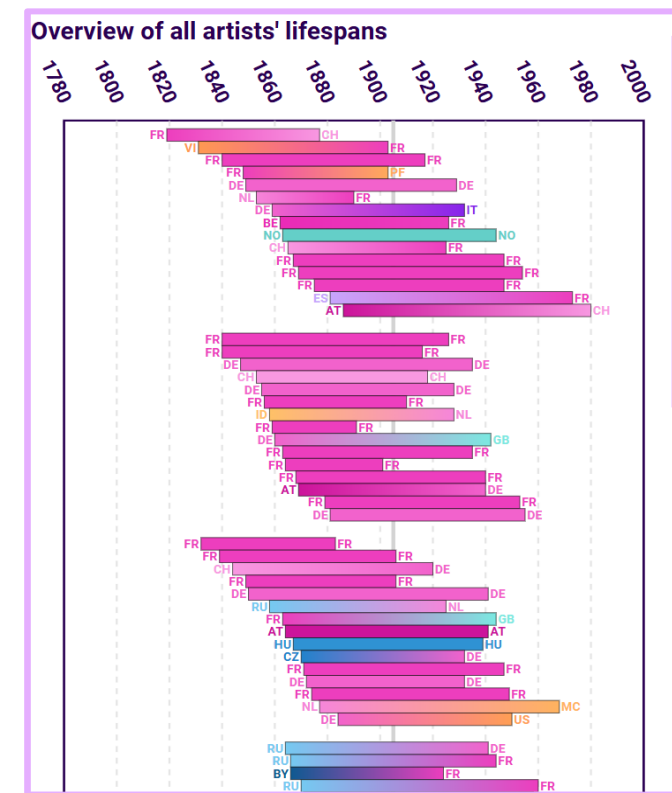


[T2] Mapping **Geographic** Influences

[R3.1] Display Artist Lifespans

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Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Temporal	The overview of the artist lifespans facilitated comparisons by quickly highlighting overlaps and notable differences in their lifespans.	6,2	6	7	7	5	6
	By hovering over artists in the overview of artist lifespans and the artists being highlighted in their respective clusters, I could see how their early/late birth or death years influenced their positioning within the cluster.	5,8	7	4	7	6	5
	The overview of the exhibitions allowed me to quickly identify periods with more or fewer exhibitions for individual artists or clusters.	6,8	7	6	7	7	7
	The exhibition timeline provided a detailed view of durations, making it easy to quickly identify long and short exhibitions.	6,6	6	7	7	6	7
	The ability to select a specific time period in the overview of the exhibitions was useful for accessing more detailed information.	6,8	7	7	7	7	6



„[Artist’s movement patterns are] **easy to see.**“ – VA

“I find it really **interesting**, because you can **directly see** the color change [if an artist migrated].” – VA



Rectangles sometimes too small for clear hovering



[T2] & [T3] Geographical and Temporal Information

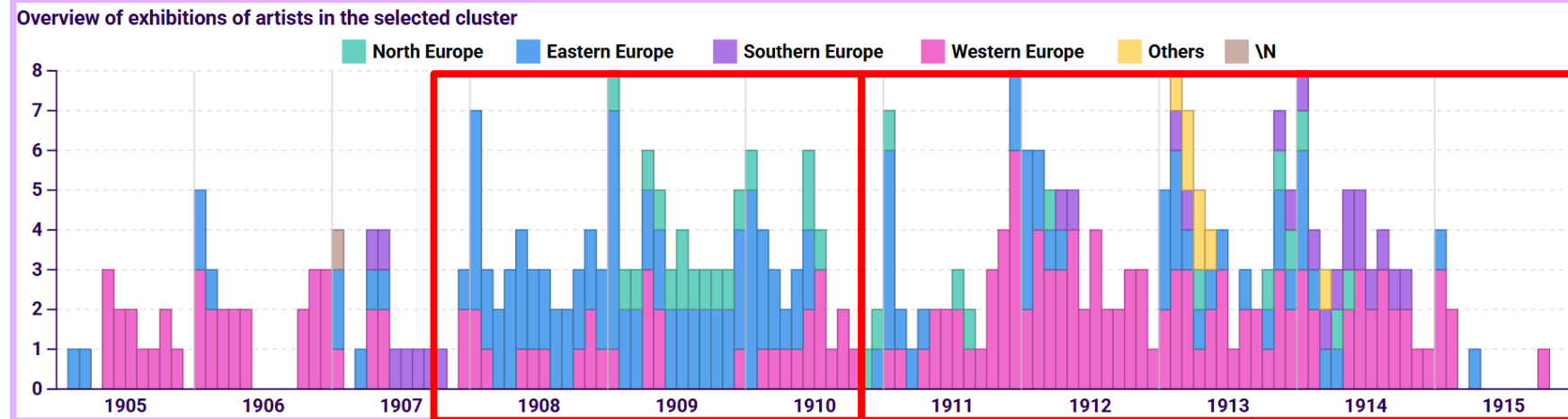
Discussion



[T2] Mapping **Geographic** Influences

[R3.2] Provide Overview of Exhibition Periods

„I really find it **interesting** that there's a **clear focus** on Eastern Europe between 1908 and 1910. And then they really **shift the focus** towards [other regions]“ – AH



Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
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[T2] & [T3] Geographical and Temporal Information

Discussion

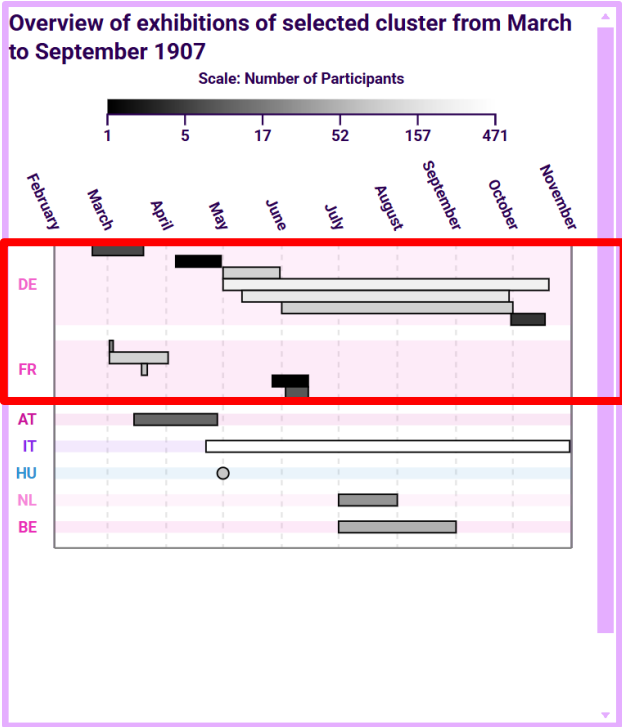


[T2] Mapping **Geographic** Influences

[R3.3] Visualize Detailed Exhibition Timelines

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	The overview of the exhibitions let me easily spot the most prominent european regions of exhibitions.	7	7	7	7	7	7
	The timeline of the exhibitions was useful in quickly identifying the most prominent countries of exhibitions.	6	6	5	7	5	7

Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Temporal	The overview of the artist lifespans facilitated comparisons by quickly highlighting overlaps and notable differences in their lifespans.	6,2	6	7	7	5	6
	By hovering over artists in the overview of artist lifespans and the artists being highlighted in their respective clusters, I could see how their early/late birth or death years influenced their positioning within the cluster.	5,8	7	4	7	6	5
	The overview of the exhibitions allowed me to quickly identify periods with more or fewer exhibitions for individual artists or clusters.	6,8	7	6	7	7	7
	The exhibition timeline provided a detailed view of durations, making it easy to quickly identify long and short exhibitions.	6,6	6	7	7	6	7
	The ability to select a specific time period in the overview of the exhibitions was useful for accessing more detailed information.	6,8	7	7	7	7	6



„The **German** exhibitions are a lot longer than the **French** exhibitions“
– AH



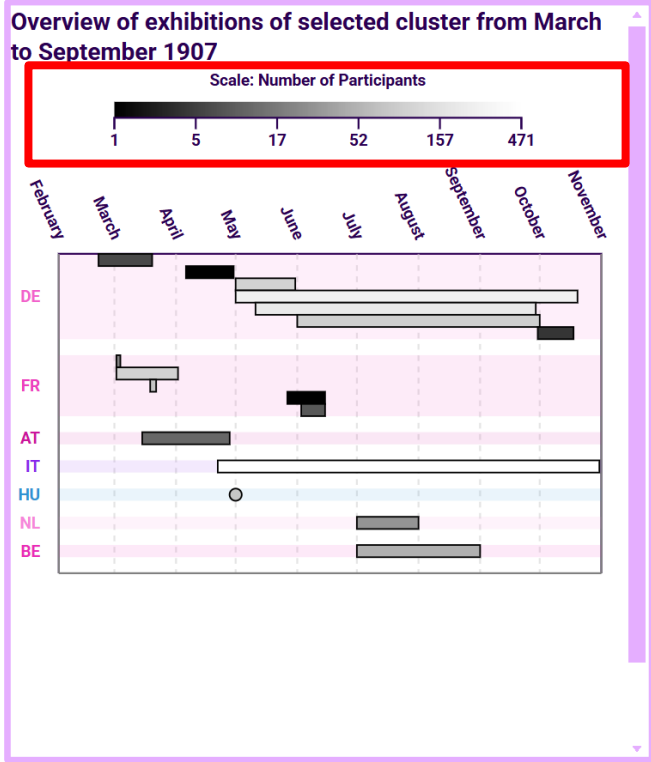
[T4] Color Scale for Highlighting Contextual Information

Discussion

[R4.2] Display Exhibition Details

Main Goal: Support distinguishing between *solo* and *group* exhibitions

Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Contextual	The tooltips in the components provided the detailed information I needed to explore the data.	6,6	7	6	7	7	6
	The color scale of the rectangles (exhibitions) in the exhibition timeline made it easy to quickly extract the number of participants.	6,6	7	5	7	7	7
	Comparing an artist's distribution of techniques to their cluster helped me understand the artist's relevance to specific techniques within that cluster.	6,2	7	6	7	4	7
	Comparing techniques among connected artists helped me identify both similarities and differences in their artistic styles, deepening my understanding of their connections.	5,8	7	7	7	2	6
	The AI-generated answers gave me useful insights into artist connections.	6,2	6	5	7	7	6



„[It is] pretty **easy to see** [exhibitions with the most/least participants]“ – AH

„You see if it’s a solo or group exhibition very **fast and quickly**“ – AS1



Due to normalization unclear number of participants for **grey colored** exhibitions



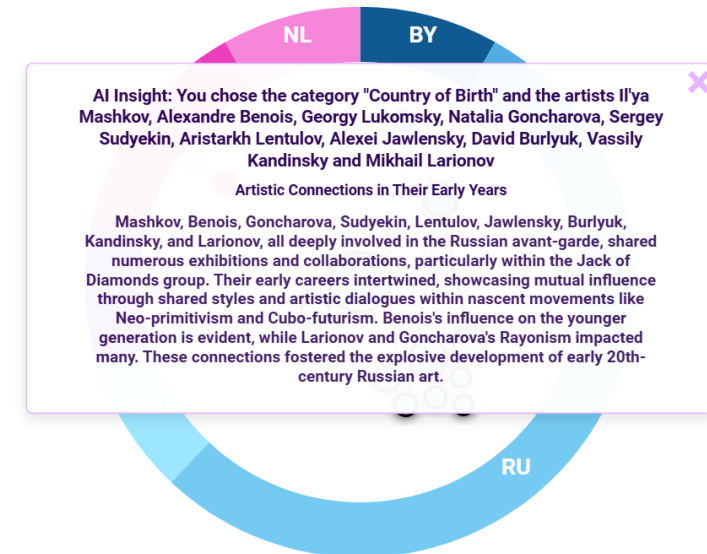
[T4] AI-Suggestions for Additional Information

Discussion

[R4.4] Support AI-Generated Insights

Category	Likert-Scale-Statement	Mean	DB	VA	AH	AS1	AS2
Contextual	The tooltips in the components provided the detailed information I needed to explore the data.	6,6	7	6	7	7	6
	The color scale of the rectangles (exhibitions) in the exhibition timeline made it easy to quickly extract the number of participants.	6,6	7	5	7	7	7
	Comparing an artist's distribution of techniques to their cluster helped me understand the artist's relevance to specific techniques within that cluster.	6,2	7	6	7	4	7
	Comparing techniques among connected artists helped me identify both similarities and differences in their artistic styles, deepening my understanding of their connections.	5,8	7	7	7	2	6
	The AI-generated answers gave me useful insights into artist connections.	6,2	6	5	7	7	6

AI suggestion on connections



AH mentioned it could **benefit both art historians and general users**

„One of the **exciting** points“ – AS2

„didn't read **any misleading information**“ – AS2, „[AI Responses seem] **plausible**“ – AS1

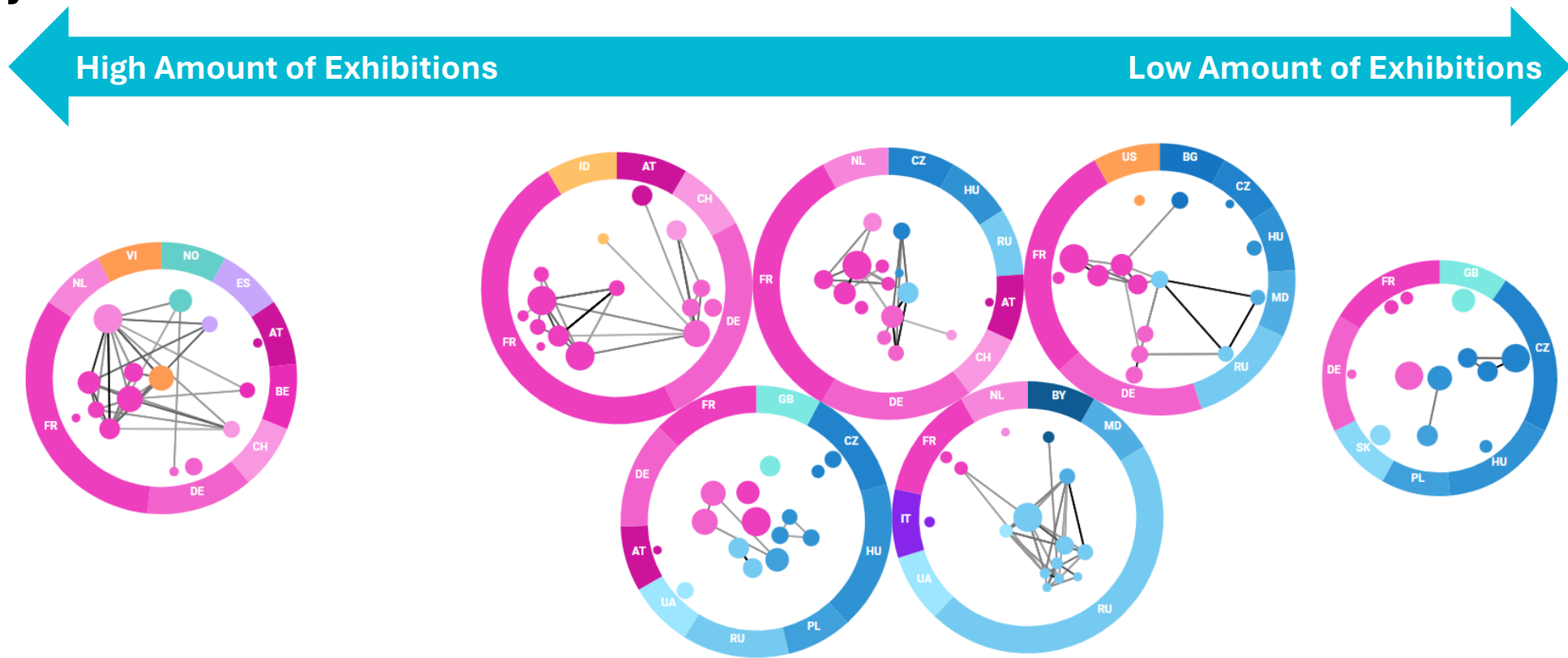


Doubts in AI: „Seems plausible, but **I don't know if it's true**“ - DB

Discussion

Discussion

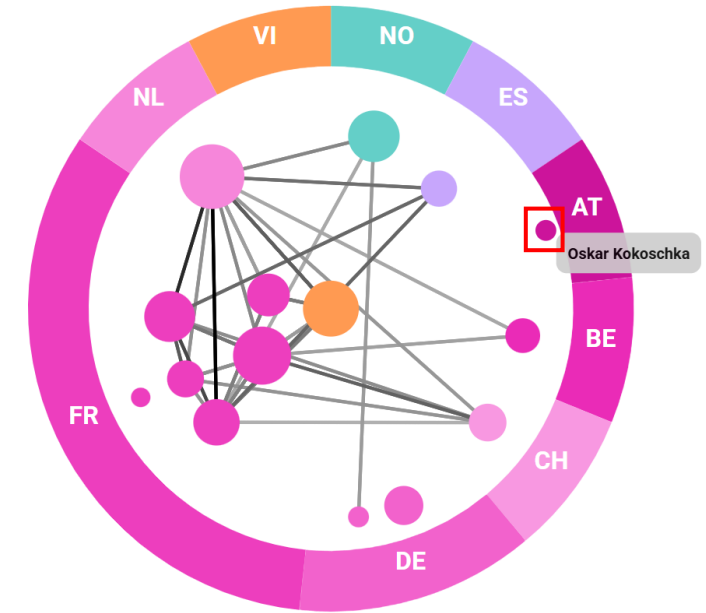
- 1. AH identified regional trend from Western Europe (left) to Eastern Europe (right) shows difference in exhibition acitivity.**



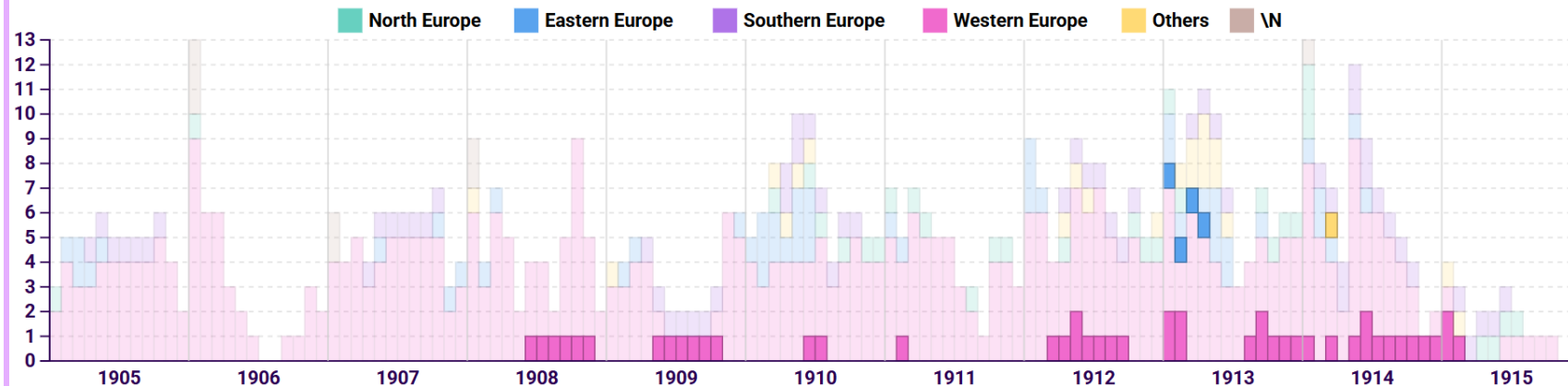
Further Findings

Discussion

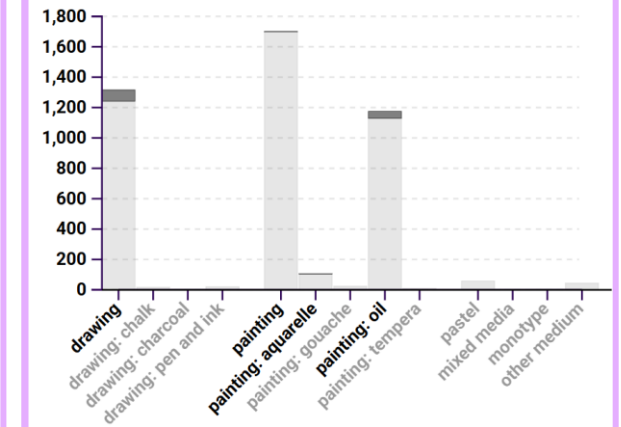
1. AH identified regional trend from Western Europe (left) to Eastern Europe (right) shows difference in exhibition acitivity.
2. Three Participants agreed that the comparison of the artist and his cluster helps to understand the artist's position in his cluster and his importance.



Exhibitions of Oskar Kokoschka



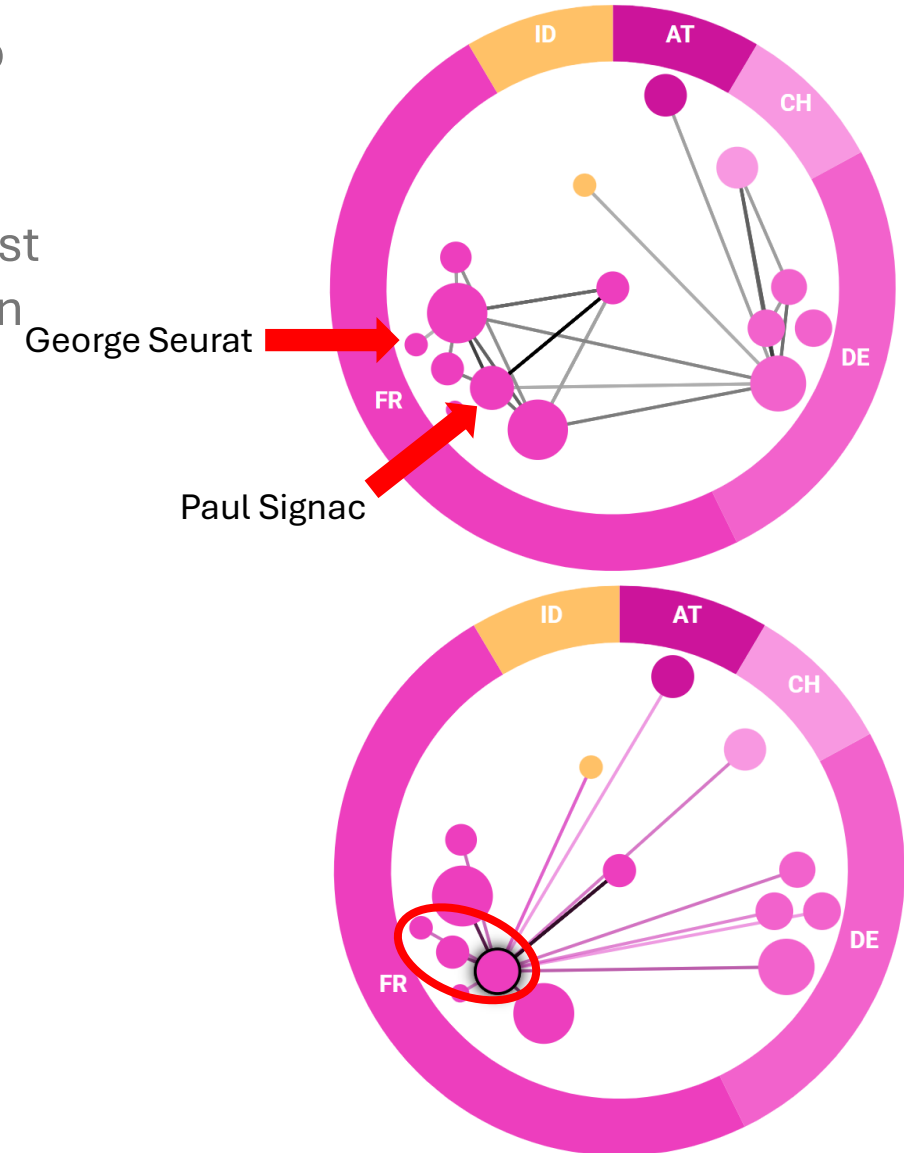
Techniques used by Oskar Kokoschka



Further Findings

Discussion

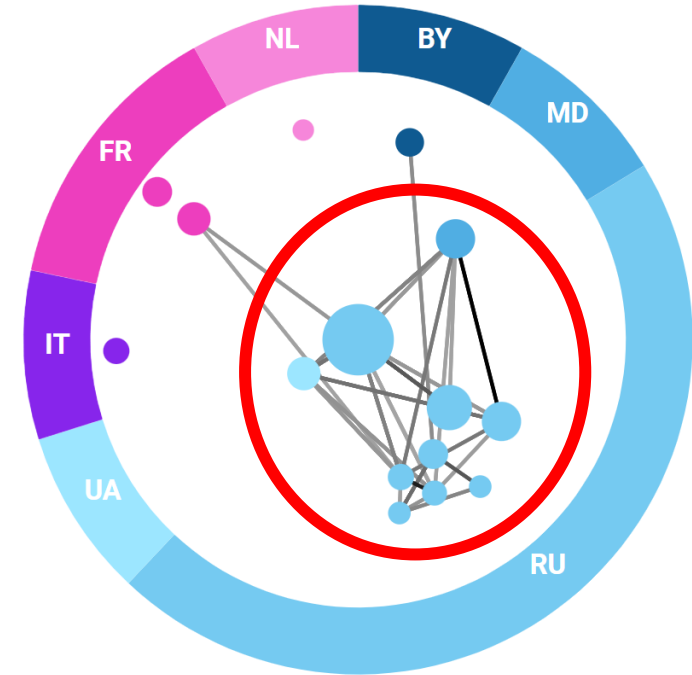
1. AH identified regional trend from Western Europe (left) to Eastern Europe (right) shows difference in exhibition activity.
2. Three Participants agreed that the comparison of the artist and his cluster helps to understand the artist's position in his cluster and his importance.
3. „I always was wondering if people who have the same stylistic tendencies [like Georges Seurat and Paul Signac] are necessarily very strongly connected, but obviously not“ – AH



Further Findings

Discussion

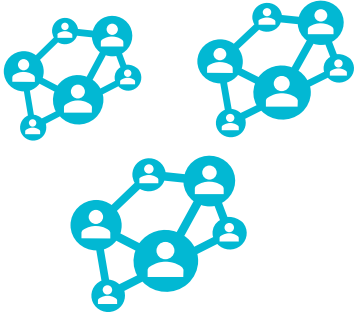
1. AH identified regional trend from Western Europe (left) to Eastern Europe (right) shows difference in exhibition activity.
2. Three Participants agreed that the comparison of the artist and his cluster helps to understand the artist's position in his cluster and his importance.
3. „I always was wondering if people who have the same stylistic tendencies [like Georges Seurat and Paul Signac] are necessarily very strongly connected, but obviously not“ – AH
4. **AH recognized a cluster as group of artists of the „Russian avant-garde“.**



*“these 3 artists are from the **same regions**, and they form [...] **a small cluster**” – VA*

*“the **color** really helped, but also the **positioning**” – AH*

Limitations



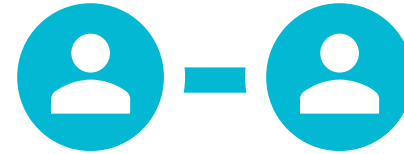
Spectral Clustering



AI-Feature only as good as used AI



Risk of False Inference



Connections between Artists

- Intra-community edges clutter and crossings
- No direct inter-community edges



Data Limitation

- Bias on western european data?
- Techniques
- Focused on Exhibitions, not Artists

Future Work

Contextualizing

by Adding Visual Information

Vincent Van Gogh
(**Impressionism**)



Vassily Kandinsky
(**Expressionism**)



Expand Application Area

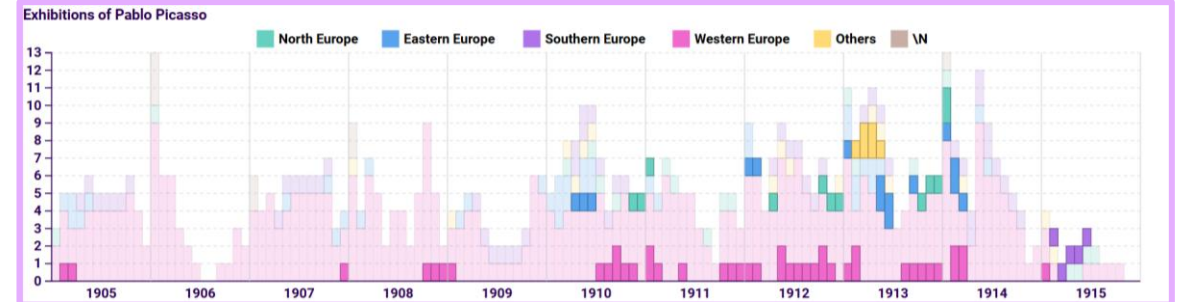
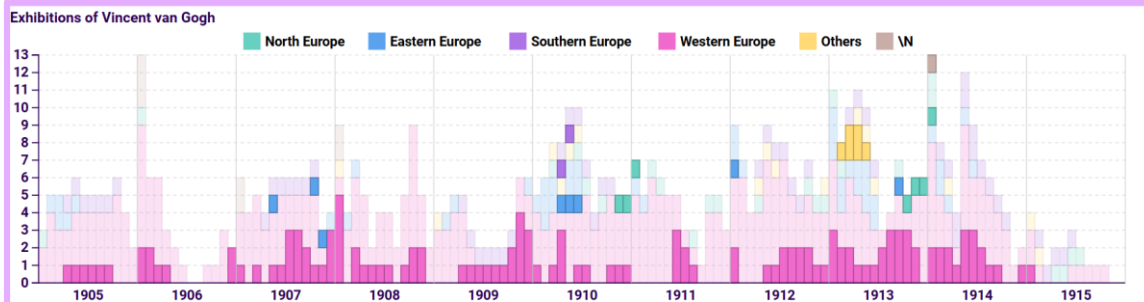
- Different databases
- E.g. ArtLas
 - international, broader time period
- Digital humanities
 - Composer Collaborations
 - Performance Networks in Theater



<https://artlas.huma-num.fr/map/#/result?collection=all&value=&type=exhibition&page=1>

Side-by-Side-View

- Enhance direct comparisons between artists (e.g. Exhibition Overview)



Conclusion

Interactively **generate insights** about artist connections across **different dimensions**:



Cluster & Network structure, Interactivity



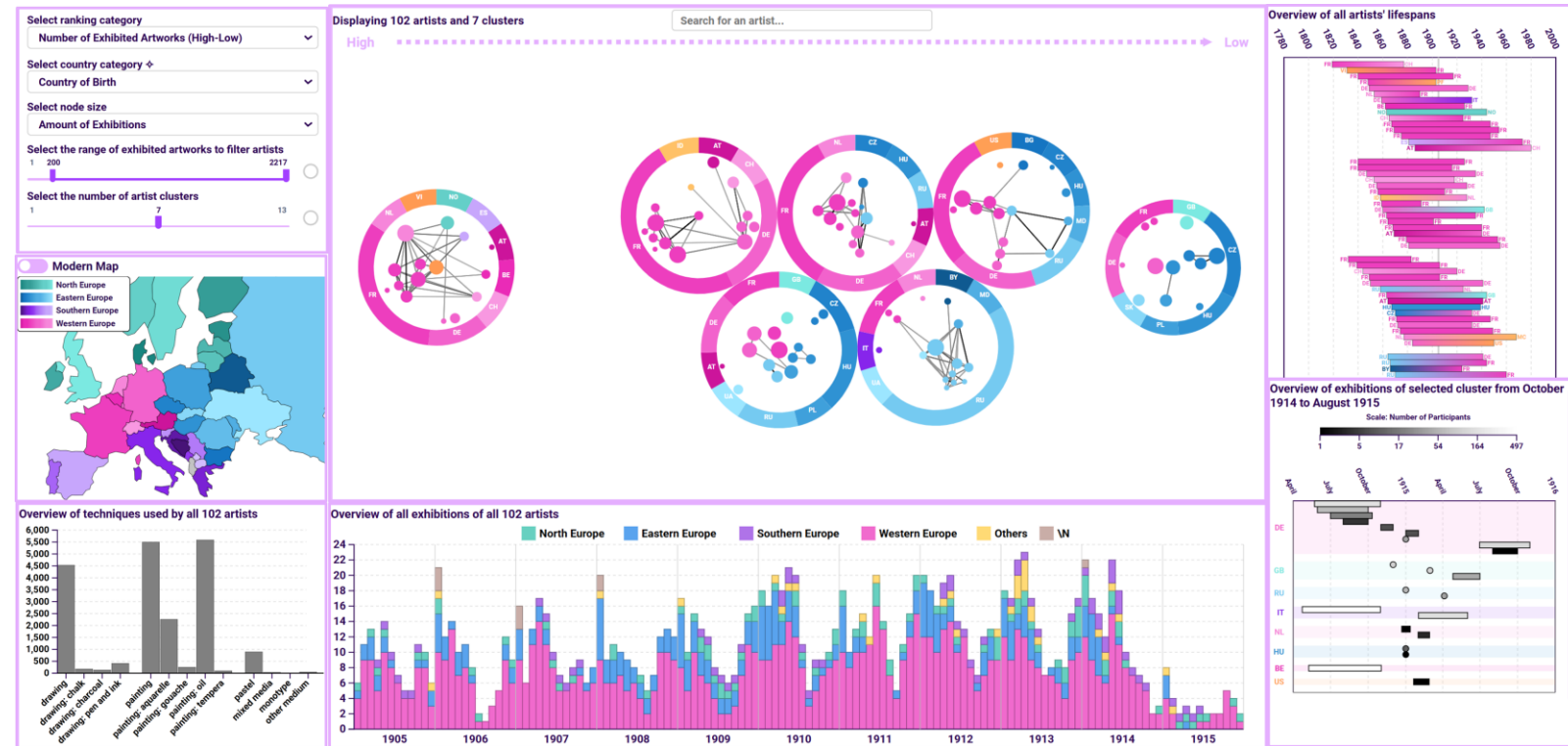
Regional color scheme



Overview first, Details on Demand



AI-generated Insights



Support art historians to uncover connections that would be difficult to discern using traditional methods alone